



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 17 Optimization III: Convex Functions (Continued)

Outline

- Operations that Preserving Convexity
 - Affine transformation
 - Pointwise maximum
 - Non-negative weighted sum

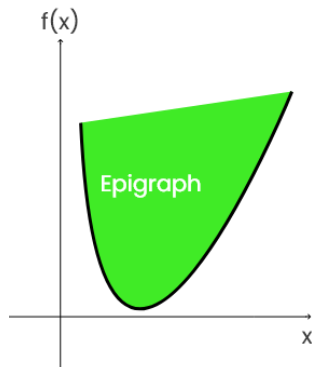
Convex Function

1. *Algebraic characterization:*

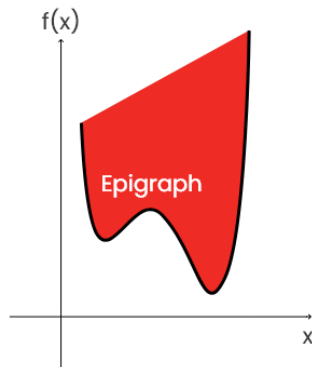
$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2)$$

that is, value of a convex combination of points $\leq$ convex combination of values at these points

2. *Geometric interpretation:* the epigraph (i.e., graph on top of the function) is a convex set



A convex function



Not a convex function

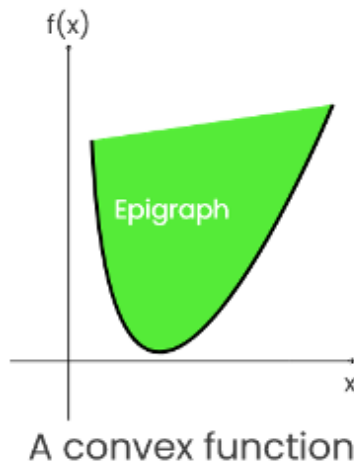
Convex Function, Convex Set, and Epigraph

The **epigraph** of f , denoted as:

$$\mathbf{epi}(f) = \{(x, t) \mid t \geq f(x)\}$$

is the set of points lying **above** the function graph.

Theorem: A function f is convex if and only if its epigraph is a convex set.



Operations that Preserving Convexity

I. Affine Transformation (Scaling and Shifting)

- If f is convex, then $g(\mathbf{x}) = f(a\mathbf{x} + \mathbf{b}) - c$ is also convex
 - a : scaling
 - $\mathbf{b}$: horizontal shift
 - c : vertical shift

I. Affine Transformation (Scaling and Shifting)

Proof. By algebraic definition, we need to show that

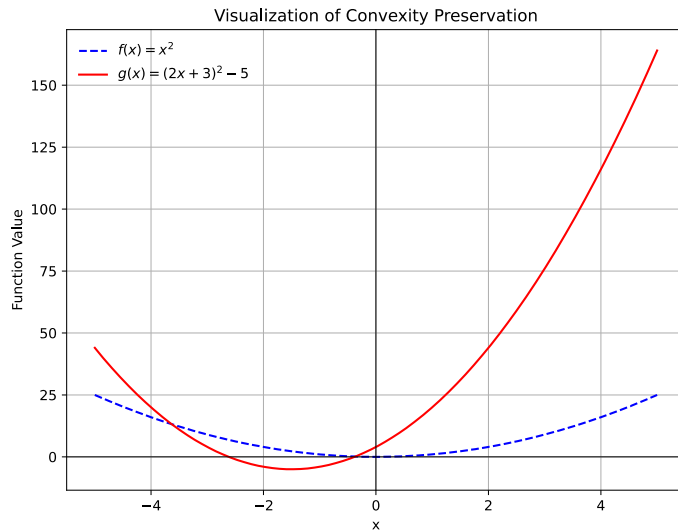
$$g(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda g(\mathbf{x}) + (1 - \lambda) g(\mathbf{y}).$$

Indeed,

$$\begin{aligned} & g(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \\ &= f[a(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) + \mathbf{b}] - c \\ &= f[\lambda (a\mathbf{x} + \mathbf{b}) + (1 - \lambda)(a\mathbf{y} + \mathbf{b})] - c \\ &\leq \lambda f(a\mathbf{x} + \mathbf{b}) + (1 - \lambda)f(a\mathbf{y} + \mathbf{b}) - c && \text{[convexity of } f\text{]} \\ &= \lambda (f(a\mathbf{x} + \mathbf{b}) - c) + (1 - \lambda) (f(a\mathbf{y} + \mathbf{b}) - c) \\ &= \lambda g(\mathbf{x}) + (1 - \lambda)g(\mathbf{y}) \end{aligned}$$

Exercise

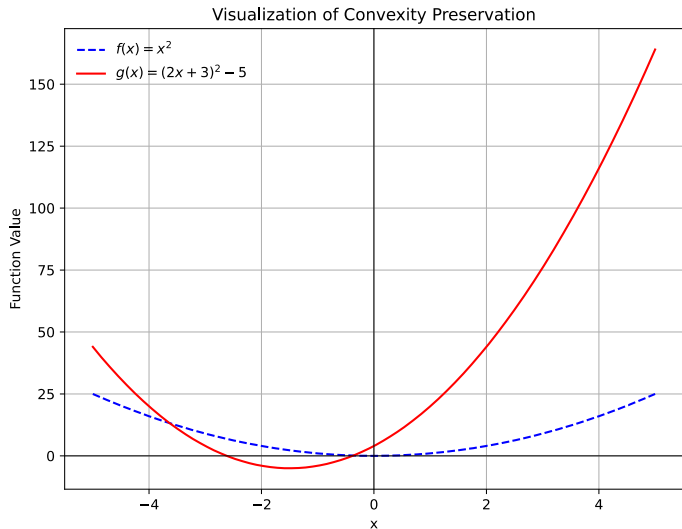
Using the convexity of $f(x) = x^2$ to show that $g(x) = (2x + 3)^2 - 5$ is also convex.



Exercise

Using the convexity of $f(x) = x^2$ to show that $g(x) = (2x + 3)^2 - 5$ is also convex.

Proof. Consider the linear transformation defined by $a = 2$, $b = 3$, $c = 5$.



II. Pointwise Maximum

- If $f_1, \dots, f_m$ are convex, then the function

$$f(\mathbf{x}) = \max\{f_1(\mathbf{x}), \dots, f_m(\mathbf{x})\}$$

is also convex.

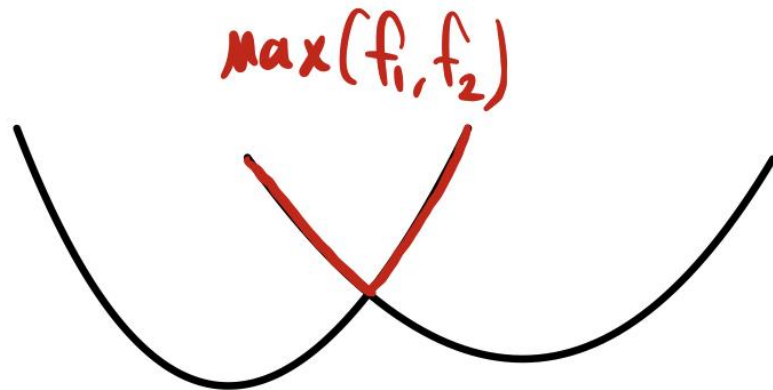
- It holds for the maximum of even *infinitely many* functions

II. Pointwise Maximum

Proof. (Geometric)

The epigraph of f is the intersection of epigraphs of $f_1, \dots, f_m$. Since $f_1, \dots, f_m$ are convex, their epigraphs are convex sets.

Since the intersection of convex sets is convex (previous lecture), the epigraph of f is a convex set, hence f is convex.



II. Pointwise Maximum

Alternative Proof. (Algebraic)

For any x, y , and $\lambda \in [0, 1]$, it holds that

$$\begin{aligned} & f(\lambda x + (1 - \lambda)y) \\ &= f_j(\lambda x + (1 - \lambda)y) && \text{for some } j \in \{1, \dots, m\} \\ &\leq \lambda f_j(x) + (1 - \lambda)f_j(y) && [\text{convexity of } f_j] \\ &\leq \lambda \max\{f_1(x), \dots, f_m(x)\} + (1 - \lambda) \max\{f_1(y), \dots, f_m(y)\} \\ &= \lambda f(x) + (1 - \lambda)f(y) \end{aligned}$$

Exercise

- Are the following functions convex? Why?
 - $f(x) = \max\{x^2, |x|\}$
 - $f(x) = \max\{2x + 3, -x + 5\}$
 - $f(x) = \max\{x_1^2 + x_2^2, |x_1| + |x_2|\}$

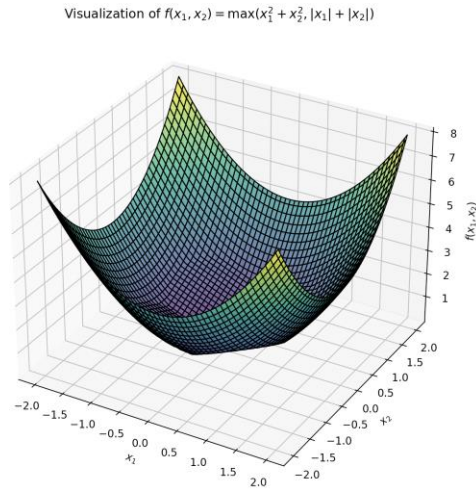
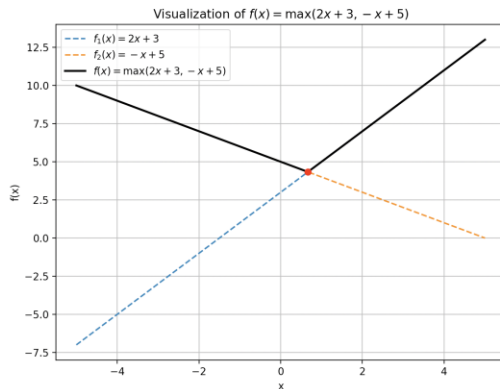
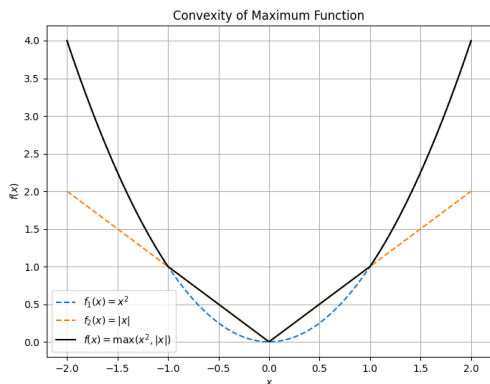
Exercise

- Are the following functions convex? Why?
 - $f(x) = \max\{x^2, |x|\}$
 - $f(x) = \max\{2x + 3, -x + 5\}$
 - $f(x) = \max\{x_1^2 + x_2^2, |x_1| + |x_2|\}$
- Yes, they are all maximum of convex functions

Exercise

- Are the following functions convex? Why?

- $f(x) = \max\{x^2, |x|\}$
- $f(x) = \max\{2x + 3, -x + 5\}$
- $f(x) = \max\{x_1^2 + x_2^2, |x_1| + |x_2|\}$



Example

Show that

$$f(x) = \sup_{y \in C} \|x - y\|_2$$

is convex in x .

Here

$$\|x - y\|_2 = \sqrt{\sum_i^n (x_i - y_i)^2}$$

is the Euclidean distance. Thereby, $f(x)$ looks for the farthest point from x within the set C .

Example

Proof.

- Since $f(\mathbf{x})$ is supremum of functions indexed by $\mathbf{y}$:

$$g_{\mathbf{y}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{y}\|_2$$

It suffices to show $g_{\mathbf{y}}(\mathbf{x})$ is convex in $\mathbf{x}$ for any fixed $\mathbf{y}$.

- Furthermore, since the shift preserves convexity, it suffices to show that $g_0(\mathbf{x}) = \|\mathbf{x}\|_2$ is convex.
- But this follows from *triangle inequality*

$$\|\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2\|_2 \leq \lambda \|\mathbf{x}_1\|_2 + (1 - \lambda) \|\mathbf{x}_2\|_2$$

Exercise

- A seller wants to improve the quality of a product.
- If the seller chooses a quality level q , the **cost of improving the quality** is $kC(q)$

where k is a cost factor.

- The **revenue** from selling the product at quality q is $R(q)$
- *Suppose the seller chooses the best quality q that gives the highest profit.*
- The seller's **profit** is given by:

$$\pi(k) = \max_{q \in Q} [R(q) - kC(q)],$$

here Q is a subset of the real line

- *Question: is $\pi(k)$ convex in k ?*

Exercise

- The function

$$f_q(k) = R(q) - kC(q)$$

is linear and thus convex in k . This function is indexed by q .

- $\pi(k)$ can be expressed as

$$\pi(k) = \max_{q \in Q} f_q(k)$$

hence the pointwise maximum of convex functions, and thus it is convex.

- This holds even for an infinite index set Q , e.g. an interval in the real line

III. Non-negative Weighted Sums

- If $f_1, f_2, \dots, f_n$ are convex functions, and $w_1, w_2, \dots, w_n \geq 0$, then the function

$$f(x) = w_1 f_1(x) + w_2 f_2(x) + \dots + w_n f_n(x)$$

is convex.

- *Useful in financial portfolio selection*

III. Non-negative Weighted Sums

Proof. In the previous lecture, we have shown that

1) If f is convex, then f/c is convex for any $c > 0$

2) If f_1, f_2 are convex, then $f_1 + f_2$ are convex

Using property (1), $w_1 f_1, w_2 f_2, \dots, w_n f_n$ are convex functions.

Using property (2), their sum is also a convex function.

Exercise

Is $f(x) = 0.7x^2 + 0.3|x|$ a convex function? Why?

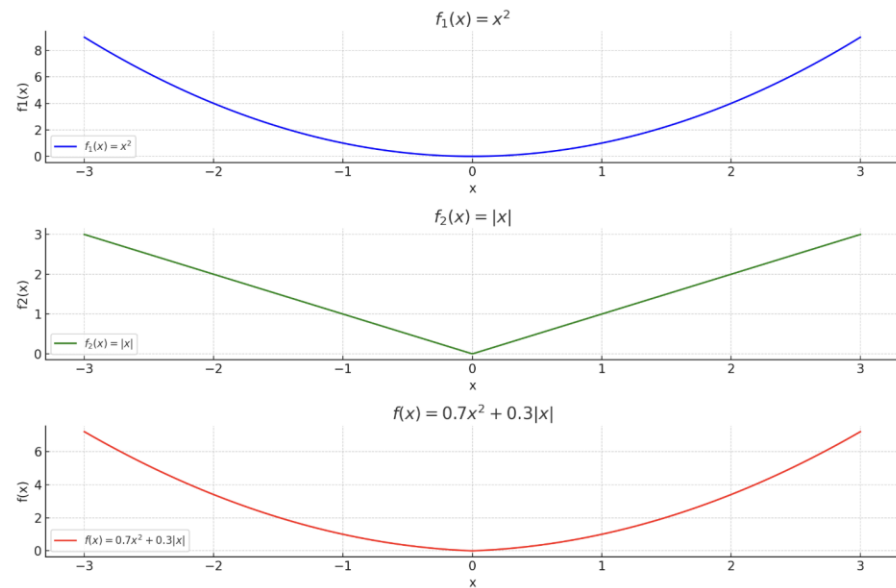
Exercise

Is $f(x) = 0.7x^2 + 0.3|x|$ a convex function? Why?

Yes.

Both $f_1(x) = x^2$, $f_2(x) = |x|$ are convex functions.

The nonnegative weighted sum of convex functions with weights (0.7, 0.3) is also convex.



Exercise

Prove that the function $f(x) = 2x^2 + 3|x|$ is convex.

Exercise

Prove that the function $f(x) = 2x^2 + 3|x|$ is convex.

Proof.

- We know x^2 and $|x|$ are convex, e.g., by looking at their epigraphs.
- Multiplication by a positive constant preserves convexity $\rightarrow 2x^2$ and $3|x|$ are convex.
- Summation preserves convexity $\rightarrow f(x)$ is convex.

III. Non-negative Weighted Sums

- The result holds when the weighted sum is over an infinite index set
- In this case, the weights can be viewed as a probability distribution
- If $f(x, y)$ is convex in x for each fixed $y \in \mathcal{C}$, and $w(y)$ is a probability density function, then the function $g(x)$ defined by:

$$g(x) = E_{y \sim w}[f(x, y)] = \int_{\mathcal{C}} w(y) f(x, y) dy$$

is convex in x , provided the function is integrable.

Summary

- Operations that Preserving Convexity
 - Affine transformation
 - Scaling & horizontal shift of variables and vertical shift of function values
 - Pointwise maximum
 - Can be max over an infinite set
 - Non-negative weighted sum
 - Multiplication by a positive constant and summation
 - Holds for infinite sums (i.e. integral)

Midterm Survey:



Your feedback is like the secret ingredient to make this course better! 🍀 Thank you for helping me improve the quality!